ACADEMIC PROJECT

Simple Calculator Application

An Android Studio project built with Java and XML



ANDROID STUDIO PROJECT

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Project at a glance

A small, practical app for entering and evaluating everyday mathematical expressions.

What I built

I developed an Android calculator that takes input from an on-screen keypad and displays the result in the same interface.

The focus was on a clear layout, responsive buttons and correct handling of common expressions.

PLATFORM	**Android**
IDE	**Android Studio**
LANGUAGE	**Java**
LAYOUT	**XML**

What I wanted to practise

The project helped me connect interface design with application logic.

01 Android UI

Build the calculator screen using XML and Material Buttons.

02 Java events

Connect keypad buttons to click-handling methods.

03 Problem solving

Evaluate expressions while respecting operator precedence.

04 Usability

Keep common calculations simple and easy to access.

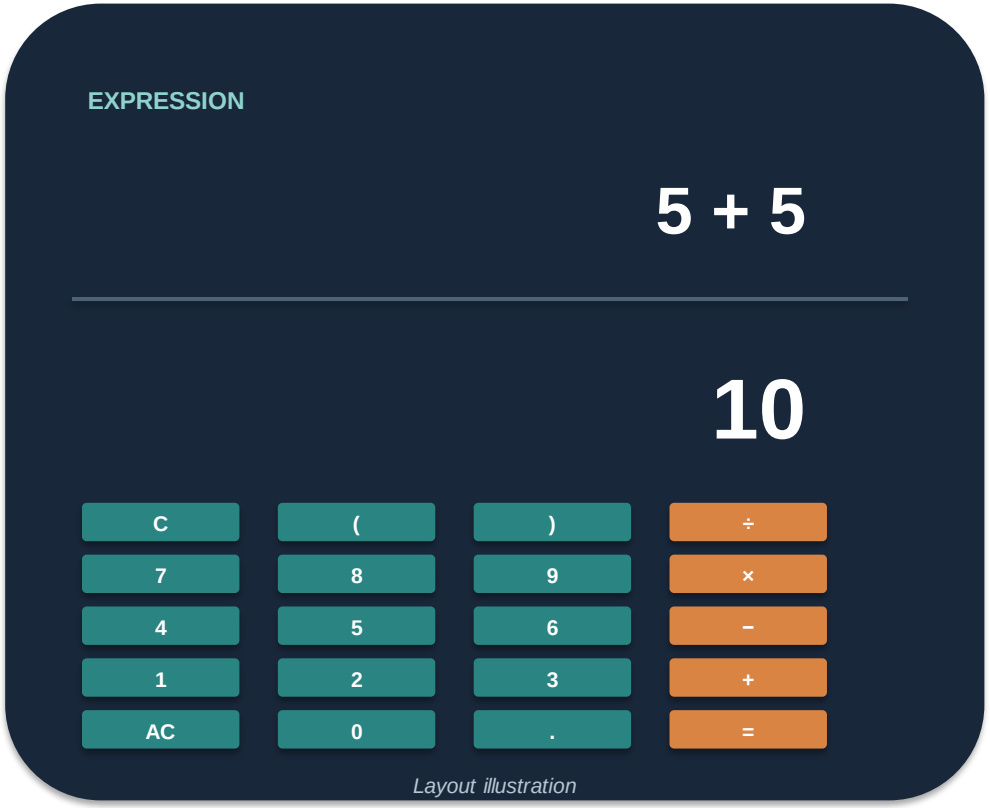
Technologies used

Each tool had a specific role in the app.

01	**Android Studio**	Created, ran and tested the Android project.
02	**Java**	Handled button clicks and calculation logic.
03	**XML**	Defined the screen layout and button arrangement.
04	**MaterialButton**	Provided the calculator's rounded keypad buttons.
05	**TextView**	Displayed the entered expression and result.

How the screen is organised

The layout separates the display area from the keypad.



Layout illustration

Key layout points

- Two TextViews show the current expression and its result.
- The keypad is arranged in five rows.
- Rounded buttons keep the controls visually consistent.
- Operators and the equals button are highlighted.

What the app can do

The calculator supports the operations and controls implemented in the project.

Basic arithmetic

Addition, subtraction, multiplication and division.

Brackets

Use parentheses to group parts of an expression.

Decimal input

Enter values such as 2.5 or 3.75.

C

Delete the last character entered.

AC

Clear the expression and reset the result.

Error handling

Show “Err” for invalid calculations or division by zero.

How a calculation is processed

The keypad input moves through the Java logic before the answer is shown.

1 → 2 → 3 → 4

Press a button

The app reads the button text.

Build expression

The input is added to solutionTv.

Evaluate

getResult() calls the expression parser.

Show result

resultTv displays the answer.

Example 5 + 5 → 10

MainActivity and calculation logic

The Java class connects the controls with the calculator's behaviour.

```
public class MainActivity extends AppCompatActivity
        implements View.OnClickListener {

    private TextView resultTv;
    private TextView solutionTv;

    private void setClickListener(
            MaterialButton... buttons) {
        for (MaterialButton button : buttons) {
            button.setOnClickListener(this);
        }
    }
}
```

Why this structure?

- findViewById() connects the layout elements to Java.
- One click listener handles the keypad buttons.
- getResult() passes the expression to the parser.
- Separating UI and logic makes the code easier to follow.

How expressions are evaluated

The parser separates operators by precedence instead of calculating strictly from left to right.

parseExpression() — **+ and −**

Handles addition and subtraction.

parseTerm() — **× and ÷**

Handles multiplication and division first.

parseFactor() — **Numbers, decimals, brackets and signs**

Reads individual values and grouped expressions.

Example

10 + 6 × 2

10 + 12

= 22

Division by zero and invalid expressions return “Err”.

Checks performed

These examples cover normal input, decimals, brackets and the clear controls.

INPUT / ACTION	EXPECTED OUTPUT	WHAT IT CHECKS
5 + 5	10	Basic arithmetic
2.5 × 4	10	Decimal input
(8 + 2) ÷ 2	5	Brackets
10 ÷ 0	Err	Division by zero
AC	0	Clear all input
C	Last character removed	Delete one character

The app was checked with these sample expressions and control actions.

Actual output

The app was run in the Android Studio emulator.

- The expression 5 + 5 is entered using the keypad.
- The result 10 appears on the display.
- The screenshot is from the project emulator run.

Observed result: 5 + 5 = 10



Original emulator screenshot

What I learned

The project gave me practice with the basic parts of an Android application.

Skills practised

- Designing an interface with XML.
- Handling input through Java click events.
- Writing an expression parser.
- Testing common cases and error conditions.

Possible next steps

- Add calculation history.
- Introduce themes or dark mode.
- Add scientific functions.
- Improve validation for longer expressions.

THANK YOU

Simple, practical, and built to learn.



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